

1) What is the Priority among (*, /, %), (+, -) and (=) C Operators.?

- A) (*, /, %) > (+, -) < (=)
- B) (*, /, %) < (+, -) < (=)
- C) (*, /, %) > (+, -) > (=)
- D) (*, /, %) < (+, -) (+, -) == (=)

2) What is the output of the C statement.?

```
int main()
{
    int a=0;
    a = 4 + 4/2*5 + 20;
    printf("%d", a);

    return 0;
}
```

- A) 40
- B) 4
- C) 34
- D) 54

3) What is the output of the C Program.?

```
int main()
{
    int a=0;
    a = 10 + 5 * 2 * 8 / 2 + 4;
    printf("%d", a);

    return 0;
}
```

- A) 124
- B) 54
- C) 23
- D) 404

4) What is the output of the C Program.?

```
int main()
{
    int a=0;
    a = 4 + 5/2*10 + 5;
    printf("%d", a);

    return 0;
}
```

- A) 29
- B) 5
- C) 4
- D) 34

5) What is the output of the C Program.?

```
int main()
{
    int a=0;
    a = 10 + 2 * 12 /(3*2) + 5;
    printf("%d", a);

    return 0;
}
```

- A) 31
- B) 19
- C) 11
- D) 29

6) What is the output of C Program with functions.?

```
void show();

int main()
{
    show();
    printf("ARGENTINA ");
    return 0;
}

void show()
{
    printf("AFRICA ");
}
```

- A) ARGENTINA AFRICA
- B) AFRICA ARGENTINA
- C) ARGENTINA
- D) Compiler error

7) What is the output of C Program with functions.?

```
int main()
{
    show();
    printf("BANK ");
    return 0;
}

void show()
{
    printf("CURRENCY ");
}
```

- A) CURRENCY BANK
- B) BANK CURRENCY
- C) BANK
- D) Compiler error

8) How many values can a C Function return at a time.?

- A) Only One Value
- B) Maximum of two values
- C) Maximum of three values
- D) Maximum of 8 values

9) What is the output of a C program with functions.?

```
void show();

void main()
{
    show();
    printf("RAINBOW ");

    return;
}

void show()
{
    printf("COLOURS ");
}
```

- A) RAINBOW COLOURS
- B) COLOURS RAINBOW
- C) COLOURS
- D) Compiler error

10) What is the output of C Program.?

```
void show();

void main()
{
    printf("PISTA ");
    show();
}

void show()
{
    printf("CACHEW ");
    return 10;
}
```

- A) PISTA CACHEW
- B) CASHEW PISTA
- C) PISTA CASHEW with compiler warning
- D) Compiler error

11) How do you convert this char array to string.?

```
char str[]={ 'g','l','o','b','y'};
```

- A) str[5] = 0;
- B) str[5] = '\0'
- C) str[]={ 'g','l','o','b','y','\0'};
- D) All the above

12) What is the output of C Program.?

```

int main()
{
    int str[]={'g','l','o','b','y'};
    printf("A%c ",str);
    printf("A%s ",str);
    printf("A%c ",str[0]);
    return 0;
}

```

- A) A A A
- B) A Ag Ag
- C) A*randomchar* Ag Ag
- D) Compiler error

13) What is the output of C Program with arrays.?

```

int main()
{
    char str[]={"C","A","T","\0"};
    printf("%s",str);
    return 0;
}

```

- A) C
- B) CAT
- C) CAT\0
- D) Compiler error

14) What is the maximum length of a C String.?

- A) 32 characters
- B) 64 characters
- C) 256 characters
- D) None of the above

15) What is the output of C program with strings.?

```
int main()
{
    char str1[]="JOHN";
    char str2[20];
    str2= str1;
    printf("%s",str2);
    return 0;
}
```

- A) JOHN
- B) J
- C) JOHN\0
- D) Compiler error

16) C is _____ type of programming language.?

- A) Object Oriented
- B) Procedural
- C) Bit level language
- D) Functional

17) What is the present C Language Standard.?

- A) C99 ISO/IEC 9899:1999
- B) C11 ISO/IEC 9899:2011
- C) C05 ISO/IEC 9899:2005
- D) C10 ISO/IEC 9899:2010

18) C language was invented to develop which Operating System.?

- A) Android
- B) Linux
- C) Ubuntu
- D) Unix

19. If MIND becomes KGLB and ARGUE becomes YPESC, then what will DIAGRAM be in that code ?

- A. BGYEPYK

B. BGYPYEK

C. GLPEYKB

D. LKBGYP

20. In the following series, what will take the place of the question mark (?)

21 33 53 83 125 _____

A. 145

B. 181

C. 141

D. 183

21. 1, 8, 4, 27, 9, _____

A. 8

B. 9

C. 64

D. 16

22. 17 : 19 :: 47 : ?

A. 53

B. 59

C. 41

D. 34

23.

Statement: Should education be given by the government free of charge ?

Arg. I: Yes, it will help in universalization of education in the country.

Arg. II: No, there will be budgetary deficit creating some new problems.

- A.** I is strong
- B.** only argument II is strong
- C.** both the arguments are strong
- D.** neither I nor II is strong

24. A man walks 5 km south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place?

- A. East
- B. South
- C. North east
- D. South west

< HIGHER MATH >

25. In how many different ways can the letters of the word "CORPORATION" be arranged so that the vowels always come together.
- a) 810 b) 1440 c) 2880 d) 50400
26. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue.
- a) $10/21$ b) $11/21$ c) $2/7$ d) $5/7$
27. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of flight is.
- a) 1 hour b) 2 hours c) 3 hours d) 4 hours
28. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?
- a) 12 days b) 15 days c) 16 days d) 18 days
29. A vendor bought toffees at 6 for a Rupee. How many for a rupee must he sell to gain 20%.
- a) 3 b) 4 c) 5 d) 6
30. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is.
- a) 2:5 b) 3:5 c) 4:5 d) 6:7

Higher Math

- 31) $\lim_{x \rightarrow 2} \frac{3x^2 - x - 10}{x^2 - 4}$
- (A) $\frac{4}{11}$ (B) 0 (C) $\frac{11}{4}$ (D) None of these
- 32) $\lim_{x \rightarrow (-1)} \frac{x^{10} + x^5 + 1}{x - 1}$
- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{3}{4}$ (D) 0

- 33) $f(x) = x$ is an example of,
 (A) Identity Function (B) Constant Function
 (C) Quadratic Function (D) Square function
- 34) One-to-One function is also called as,
 (A) Bijective (B) Injective (C) Many-to-one
 (D) None of these
- 35) Empty set is also considered as,
 A) Unknown set
 B) Universal set
 C) Infinite set
 D) Finite set
- 36) The cardinality of the power set of $\{x: x \in \mathbb{N}, x \leq 10\}$ is,
 A) 1024 (B) 1023 (C) 2048 (D) 2043
- 37) The range of the function $f(x) = 3x - 2$ is,
 A) $(-\infty, \infty)$ (B) $\mathbb{R} - \{3\}$ (C) $(-\infty, 0)$ (D) $(0, -\infty)$
- 38) $\frac{d}{dx} \left(\frac{1}{x^n} \right) =$
 (A) $n \cdot x^{n-1}$ (B) $-n \cdot x^{-n-1}$ (C) x^{n-1} (D) None of these
- 39) $\frac{d}{dx} (\cos x) =$ _____
 (A) $\sin x$ (B) $\cos x$ (C) $-\sin x$ (D) $-\cos x$
- 40) What is the probability of leap year consisting of 53 Mondays?
 (A) 0 (B) $\frac{1}{7}$ (C) $\frac{2}{7}$ (D) 1
- 41) The sum of the roots of the equation $4x^2 + 5x + 7 = 0$.
 (A) $\frac{7}{4}$ (B) $\frac{4}{7}$ (C) $\frac{5}{4}$ (D) $-\frac{5}{4}$

- 42) IF a, b, c are in AP then,
 (A) $b = a + c$ (B) $2b = a + c$
 (C) $b^2 = a + c$ (D) $2b^2 = a + c$
- 43) The 3rd term of GP is 4. Then the product of the first 5 terms is,
 (A) 4^3 (B) 4^4 (C) 3^4 (D) 4^5
- 44) IF n^{th} term of an AP is $3n - 4$, then the 10th term of an AP is,
 (A) 10 (B) 12 (C) 22 (D) 26
- 45) Events A and B are said to be mutually exclusive if:
 (A) $P(A \cup B) = P(A) + P(B)$ (B) $P(A \cap B) = P(A) \times P(B)$
 (C) $P(A \cup B) = 0$ (D) None of these.
- 46) IF Set $A = \{a, b, c\}$, Set $B = \{b, c, d\}$ then total no. of elements in $n(A \times B) =$ _____.
 (A) 6 (B) 3^3 (C) 9 (D) None of these.
- 47) $\frac{\log \sqrt{27}}{\log 27}$ is,
 (A) 1 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) 2
- 48) Last digit in $1! + 2! + 3! + 4! + 5! + \dots + 2024!$ is,
 (A) 0 (B) 1 (C) 2 (D) 3

ENGLISH

The happy man is the man who lives objectively, who has free affections and wide interests, who secures his happiness through these interests and affections and through the fact that they, in turn, make him an entity of interest and fondness to many others. To be the recipient of affection is a potent cause of happiness, but the man who demands affection is not the man upon whom it is bestowed. The man who receives liking is, speaking generally, the man who gives it. But it is useless to attempt to give it as a calculation, in the way in which one might lend money at interest, for a

calculated affection is not honest and is not felt to be so by the beneficiary. What then can a man do who is doomed because he is enclosed in self? So long as he continues to think about the causes of his unhappiness, he continues to be self-centred and therefore does not get outside it. It must be of genuine interest, not by simulated interests adopted merely as a medicine. Although this difficulty is real, there is never the less much that he can do if he has rightly diagnosed his trouble. If for instance, his woe is due to a sense of depravity, conscious or unconscious, he can first influence his conscious mind that he has no reason to feel sinful, and then proceed, to plant this rational conviction in his unconscious mind, concerning himself meanwhile with some more or less neutral activity. If he prospers in dismissing the sense of immorality, it is possible that genuine unbiased interests will arise spontaneously. If his trouble is self-pity, he can deal with it in the same manner after first persuading himself that there is nothing extraordinarily unfortunate in his circumstances. If fright is his worry, let him drill exercises designed to give bravery. Courage has been recognized from time immemorial as an important virtue, and a great part of the training of boys and young men has been devoted to producing a type of character capable of fearlessness in battle. But moral courage and intellectual courage have been much less studied. They also, however, have their technique. Admit to yourself every day at least one painful truth, you will find it quite useful. Teach yourself to feel that life would still be worth living even if you were not, as of course, you are, immeasurably superior to all your friends in virtue and in intelligence. Exercises of this kind protracted through numerous years will, at last, enable doing, might free you from the empire of fear on a very large scale.

49. What does calculated affection mean?

- a. affection which is recieved unwillingly
- b. affection which is temporary
- c. affection which is bogus and forge
- d. affection which comes as an alm

50. According to the passage, a happy man

- a. is self-engaged
- b. has free affection and wide interests
- c. is free from outward passions
- d. has no grudge

51. Which of the following statements is NOT TRUE in the context of the passage?

- (a) The happy man has wide interests.
- (b) Courage has been recognized as an important virtue.
- (c) A happy man lives objectively
- (d) moral courage and intellectual courage has been extensively studied.

52. Which of the following virtues, according to the passage, has been recognized for long as an important virtue?

- a. isolation
- b. oblation
- c. courage
- d. unassertiveness

53. Which of the following words is SIMILAR in the meaning of the word 'simulated' as used in the passage?

- (a) imitated
- (b) accommodated
- (c) trusted
- (d) directed

54. A man who is suffering from the feeling of self-pity should

- a. develop a feeling of fearlessness.
- b. persuade himself that everything is alright in his circumstances.
- c. seek affection from others.
- d. control his passions and emotions

55. Kites last year also.

- | | |
|----|------------|
| A. | did fly |
| B. | were flown |
| C. | Flew |

55. Kites last year also.

D. Flying

56. Listen, a nice song

A. is singing

B. has sung

C. was being sung

D. is being sung

57. Listen, an announcement ... to cancel all the flights due to heavy ice-fall.

A. Making

B. is making

C. is made

D. is being made

58. Listen, she is talking you.

A. For

B. In

C. About

D. At

59. Look, a new missile

A. is launched

B. is launch

C. is being launch

D. is being launched

60. Look, many students kites.

A. are fly

B. are flown

C. are flying

D. are being flown